

Arnab Mandal

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EDUCATION

Shiv Nadar University

B.Tech in Computer Science Engineering (8.59/10.0)

Noida, Delhi-NCR, India

Aug 2023 - Jul 2027

INTERNSHIPS & EXPERIENCE

Intern, Enterprise AI Platform, Fidelity International, Gurgaon

Jun 2026 - Jul 2026

- Cut **PII** redaction latency by **92%** via **AWQ**-quantized fine-tuning of Gemma 4 on **SageMaker**, reaching **96% F1**.
- Shipped document sensitivity guardrail layer for enterprise AI gateway on **15k+** monthly uploads, audited via **ELK**.
- Drove adoption for the Ponytail skill for AI code generation resulting in **22%** fewer tokens and **27%** faster generation.
- Built a DeepEval **harness** for RAG evaluation via the RAG triad; caught **3** groundedness regressions before release.

AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati

Dec 2025 - Jan 2026

- Optimized Llava 1.5 Pipeline serving **50+** clients by migrating inference to TensorRT + Triton.
- Achieved GPU utilization increase by **30%** and serving **8x** clients with no additional resources.

Founding AI Intern, Critical AI Pvt Ltd, Bengaluru

Jul 2025 - Sep 2025

- Deployed a RAG agent for a legal firm for citation-backed clause retrieval on a fully on-prem stack, cutting inference cost **30%** vs. cloud for **early adoption** of AI.
- Built a voice-primary multimodal RAG agent with an OSM-based mapping tool for geospatial queries, pitched to Indian Navy departments.

AI Intern, WESEE, Ministry of Defence, New Delhi

May 2025 - Jul 2025

- Ran **3** lecture sessions on AI infrastructure for **15** senior team leads, driving WESEE's **early AI adoption**.
- Cut SME report prep time by **90%** via deploying **3+** MCP servers for agentic SQL analytics of radar/satellite data.
- Built LLM serving for internal MoD chatbot, used by **7+** teams and handling **200+** daily queries.

PROJECTS

Open-Source Contributor at vLLM (85k★): Domino Speculative Decoding 🔄 🧠

- Implemented Inference and Distributed training support for Domino, reporting a **18%** throughput gain on Qwen3.
- Improved acceptance length by **27%** over DFlash across 7 benchmarks, including **GSM8K** and HumanEval.
- Technology Stack: Python, PyTorch, Transformers, FSDP2

Agentic Text2SQL Engine for Enterprise Healthcare Analytics Database 🔗 🔗

- Engineered sandboxed multi-agent system with self-repair retry loop, tested over warehouse with **60+** relationships.
- Achieved **90%** query execution accuracy, and validated by **CEO+CTO** for **production** adoption.
- Technology Stack: Python, LangGraph, FAISS, NetworkX, DuckDB, dbt, Airflow

AutoLinks: Distributed Semantic Internal Linking Engine for SEO 🔗 🔗

- Architected async, autoscaling Goroutine workers with Redis-backed job queue + DLQ for bursty workloads.
- Utilized LLM-as-a-judge, achieving **85+%** acceptance, with **p95** Latency of **600ms**, lowering Gini coefficient by **20%**.
- Technology Stack: Go, chi, trafalatura, Qdrant, Redis

TECHNICAL SKILLS

Languages	Python, Rust, Go, TypeScript, C++, CUDA, SQL
AI & ML	LangChain, LangGraph, LangSmith, LiteLLM, Optuna, MLflow
MLOps & DevOps	vLLM, Triton, TensorRT, Ray Serve, Kubernetes, Docker, Jenkins, Prometheus
Cloud	GCP, AWS, Azure, Terraform

ACHIEVEMENTS

- **Smart SNU Hackathon'25 Winner**: Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- **LeetCode Knight**: Achieved a **1,925 peak contest rating** and solved **450+ DSA problems**.
- **Semifinalist speaker at SMVIT Parliamentary Debate'25**